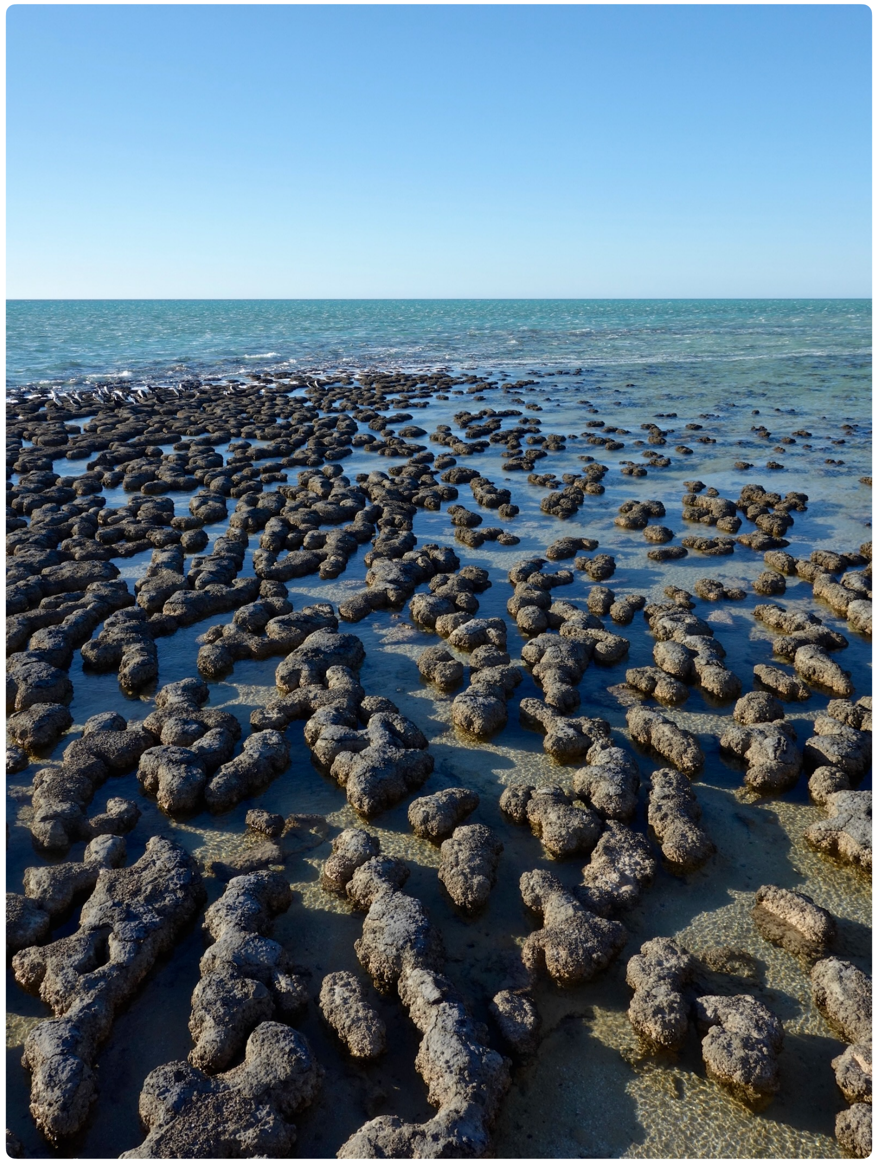
Exercises in Marine Ecological Genetics

03. *F*-statistics

and population structure I



- Calculate *F*-statistics in R
- Estimate population-specific and pairwise F_{ST}
- Visualize population structure

Martin Helmkampf

https://github.com/mhelmkampf/meg26

Today's objectives

By the end of this session, you will be able to:

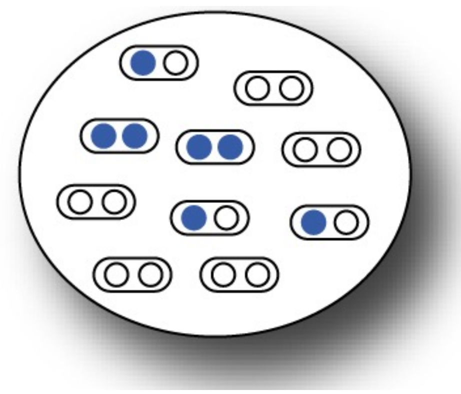
- Calculate ***F*-statistics** from microsatellite data
- Use F_{ST} to measure **genetic differentiation** (overall, population-specific and pairwise F_{ST})
- Interpret F_{ST} values and **visualize population structure** with Principal coordinates analysis
- Create basic and **improved plots** with tidyverse and ggplot

F-statistics: fixation index

Lecture

Heterozygosity

In one population



H_o = proportion of heterozygote individuals, observed heterozygosity

$H_e = 2pq = 1 - p^2 - q^2$, expected heterozygosity (assuming HW equilibrium)

$$F = \frac{H_e - H_o}{H_e}$$

Fixation index: *proportion by which heterozygosity is reduced or increased relative to the heterozygosity of a population at HW equilibrium with the same allele frequencies.*

Divided by H_e –> *proportion* (of expected heterozygosity)

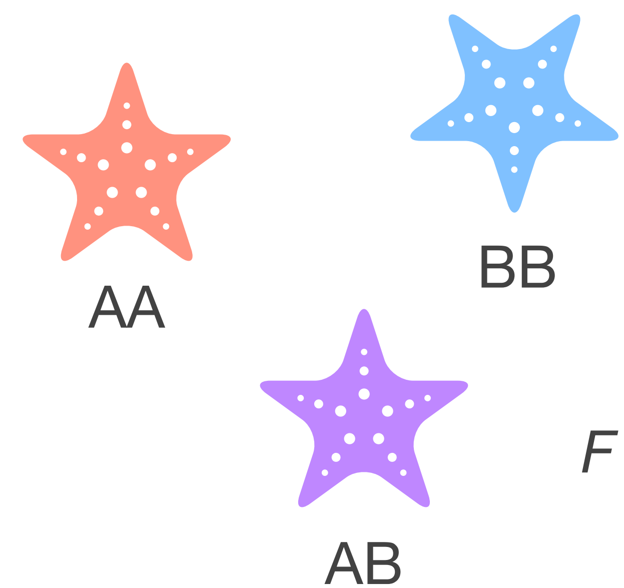
Varies between -1 and 1

F < 0: heterozygote excess

F > 0 heterozygote deficit (homozygote excess)

May be averaged over several loci –> reduces bias

May be extended to *k* alleles

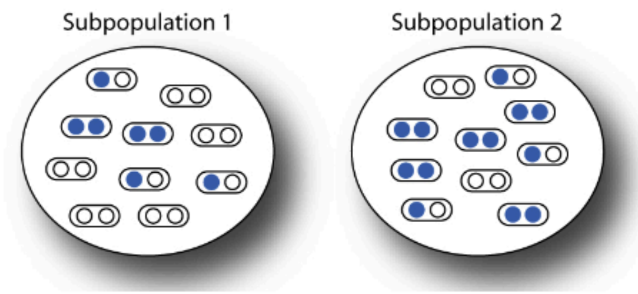


$F = 0.67$

F-statistics: multiple subpopulations

Lecture

In n subpopulations



$H_I = \frac{1}{n}\sum_{i=1}^{n}\hat{H}_i$ Mean observed heterozygosity within subpopulations

$H_S = \frac{1}{n}\sum_{i=1}^{n}2p_i q_i$ Mean expected heterozygosity within subpopulations (assuming HW within subpops)

$H_T = 2\overline{p}\overline{q}$ Expected heterozygosity of the total population (assuming HW within the total pop)

$\hat{H}_i$ observed heterozygosity in population i $p_i(q_i)$ frequency of allele A(a) in population i

$\overline{p}(\overline{q})$ mean frequency of allele A(a) n: number of subpopulations

$$F_{IS} = \frac{H_S - H_I}{H_S} \qquad F_{ST} = \frac{H_T - H_S}{H_T} \qquad F_{IT} = \frac{H_T - H_I}{H_T}$$

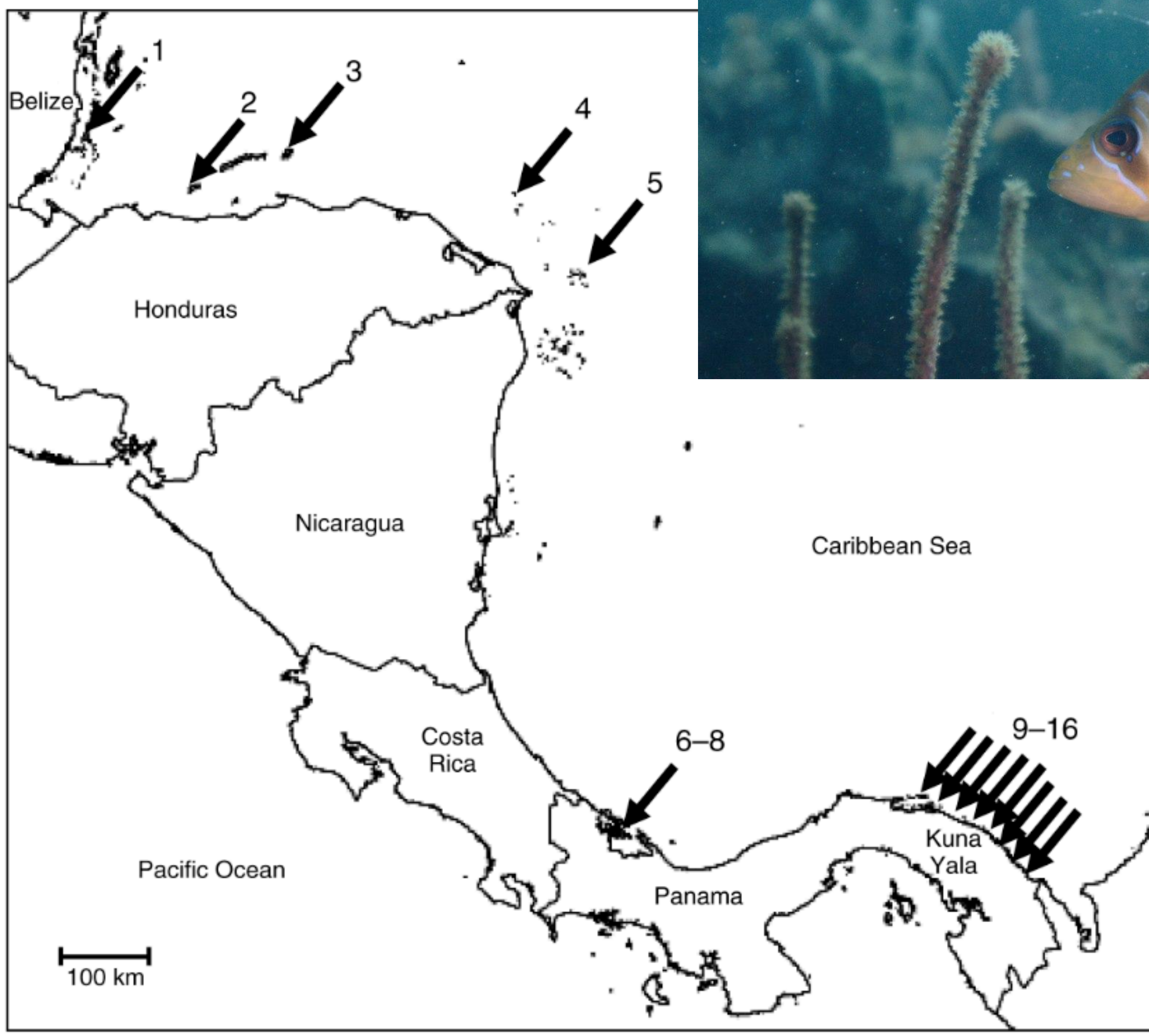
F_{IS} Average difference between observed and Hardy–Weinberg expected heterozygosity within each subpopulation (due to non-random mating)

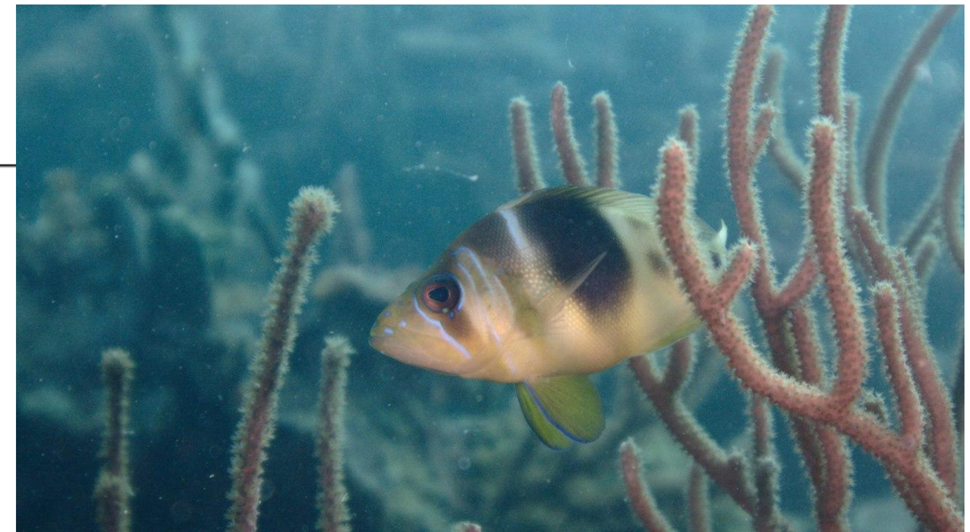
F_{ST} Reduction in heterozygosity due to subpopulation divergence of allele frequencies

F_{IT} *Combined departure from HW expected genotype frequencies due to non-random mating within subpopulations and divergence of allele frequencies among subpopulations*

Barred hamlet (*H. puella*) microsatellite dataset

Recap





Ecology, 90(11), 2009, pp. 3087–3098

Estimating dispersal from genetic isolation by distance in a coral reef fish (*Hypoplectrus puella*)

Oscar Puebla,[1,2,3] Eldredge Bermingham,[1,2] and Frédéric Guichard[2]

- 10 highly variable microsatellite loci
- 854 individuals
- 15 Caribbean locations

Genepop file format

Recap

puella_barbados.txt

```
Microsatellite genotypes of Hypoplectrus puella from Barbados
g2
gag010
h24
hyp001
hyp015
hyp018
e2
hyp008
hyp016
pam013
POP
barbados ind#735 , 203235 121127 210248 225231 126132 190192 157157 236244 194220 114128
barbados ind#736 , 205217 119119 204216 225229 126126 190192 157165 236236 216222 106144
barbados ind#737 , 203203 135147 224226 233235 126136 190198 157163 238250 202226 126160
barbados ind#738 , 211217 119121 228230 223231 126130 190190 157157 240244 220226 138160
barbados ind#739 , 205225 121125 208216 227231 132132 192192 158159 244256 192196 146146
barbados ind#740 , 217233 119121 216228 227233 126126 192198 157157 242256 194216 000000
barbados ind#741 , 203209 000000 216222 223229 126130 192198 157157 234236 188226 118130
barbados ind#742 , 203215 119119 222234 233233 130130 190190 157157 240246 182188 108126
barbados ind#743 , 203243 111121 216224 231231 126126 190192 157159 234249 188198 118126
barbados ind#744 , 203225 121135 206232 231231 126130 192198 157159 240249 182222 122156
barbados ind#745 , 203211 121123 224236 227231 126132 192192 157163 238238 182204 138148
barbados ind#746 , 227233 119123 216234 227233 126130 190190 157159 236245 186188 124142
barbados ind#747 , 203223 133141 236240 231231 126126 192198 157157 234236 194204 118126
barbados ind#748 , 203217 141145 222226 227235 126126 190192 145157 238240 182196 108108
barbados ind#749 , 205221 125147 210238 223231 126132 192198 157157 234234 188200 106148
barbados ind#750 , 203235 123123 230234 223231 126132 192198 157157 236256 188214 122136
barbados ind#751 , 203211 121143 212248 223229 126132 190196 157159 234242 214216 118128
barbados ind#752 , 213217 123169 204234 223231 126134 190198 157163 236248 193224 142148
```

Free text

Locus names

Sample / population

Alleles (000 = missing data)

Connecting to the cluster via SSH

```
### Establish SSH connection to cluster using your course account
ssh user1234@rosa.hpc.uni-oldenburg.de  # please adjust user name

### Enter password (invisible)
```

...

```
### Either: Download course materials to course account using Git (first time)
git clone https://github.com/mhelmkampf/meg26.git

### Or: Update git repository (if it already exists)
cd meg26
git pull
```

Keep following instructions in course script (03_popst.R)

F_{ST} range across organisms

Exercise 1

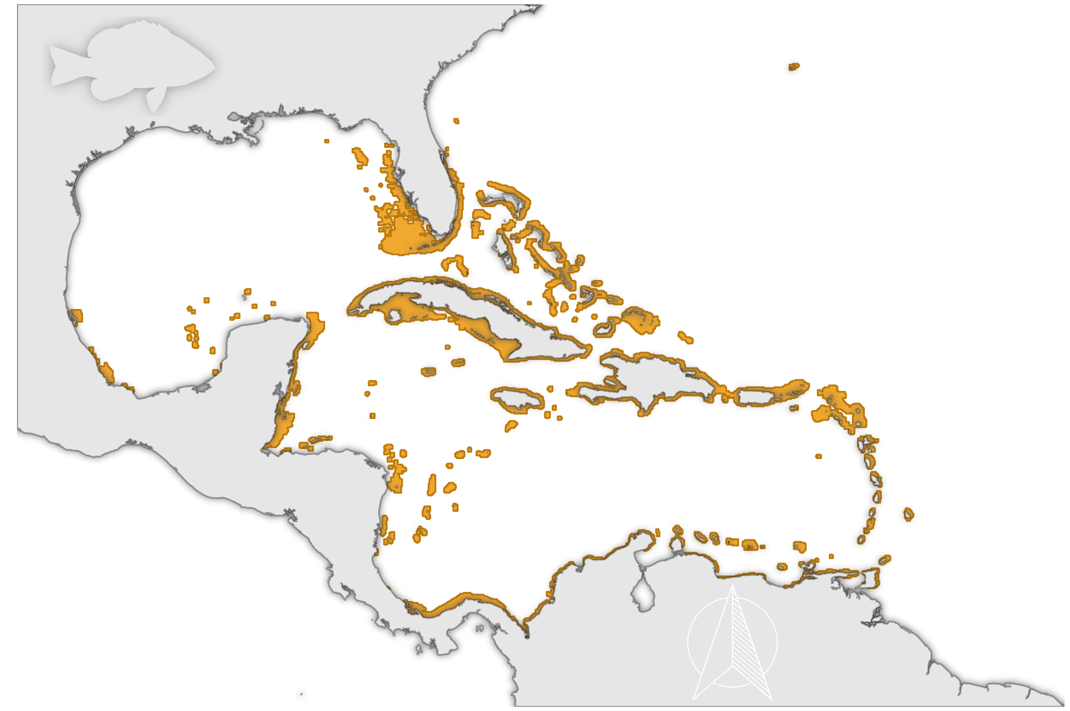
Species	$\hat{F}_{ST}$	$\widehat{N_e m}$	Reference
Amphibians			
Alytes muletansis (Mallorcan midwife toad)	0.12–0.53	1.8–0.2	Kraaijeveld-Smit et al. 2005
Birds			
Gallus gallus (broiler chicken breed)	0.19	1.0	Emara et al. 2002
Mammals			
Capreolus capreolus (roe deer)	0.097–0.146	2.2–1.4	
Homo sapiens (human)	0.03–0.05	7.8–4.6	
Microtus arvalis (common vole)	0.17	1.2	
Plants			
Arabidopsis thaliana (mouse-ear cress)	0.643	0.1	
Oryza officinalis (wild rice)	0.44	0.3	
Phlox drummondii (annual phlox)	0.17	1.2	
Prunus armeniaca (apricot)	0.32	0.5	
Fish			
Morone saxatilis (striped bass)	0.002	11.8	Brown et al. 2005
Sparisoma viride (stoplight parrotfish)	0.019	12.4	Geertjes et al. 2004
Insects			
Drosophila melanogaster (fruit fly)	0.112	2.0	Singh and Rhomberg 1987
Glossina pallidipes (tsetse fly)	0.18	1.1	Ouma et al. 2005
Heliconius charithonia (butterfly)	0.003	79.8	Kronforst and Flemming 2001
Corals			
Seriatopora hystrix	0.089–0.136	2.6–1.6	Maier et al. 200

F_{ST}	Subpopulations are …
< 0.15	very similar
0.15 – 0.25	similar
> 0.25	distinct

Hamlet species microsatellite dataset

Exercises 1–3

- Complex of 18+ species (genus *Hypoplectrus*)
- Endemic to coral reefs in wider Caribbean
- Little ecological differentiation
- Differ strongly by color pattern
- Reproductively isolated by assortative mating



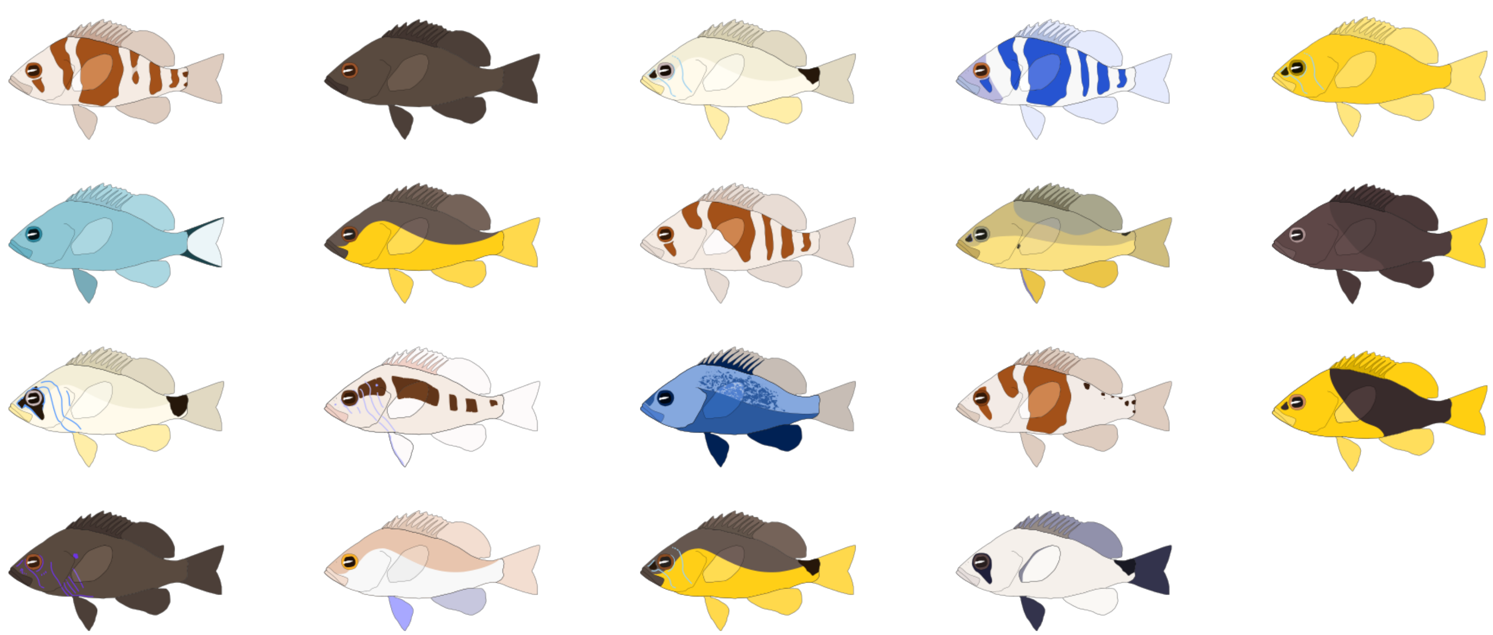


Photo by Allison & Carlos Estape

Illustrations and map by Kosmas Hench, data from biogeodb.stri.si.edu

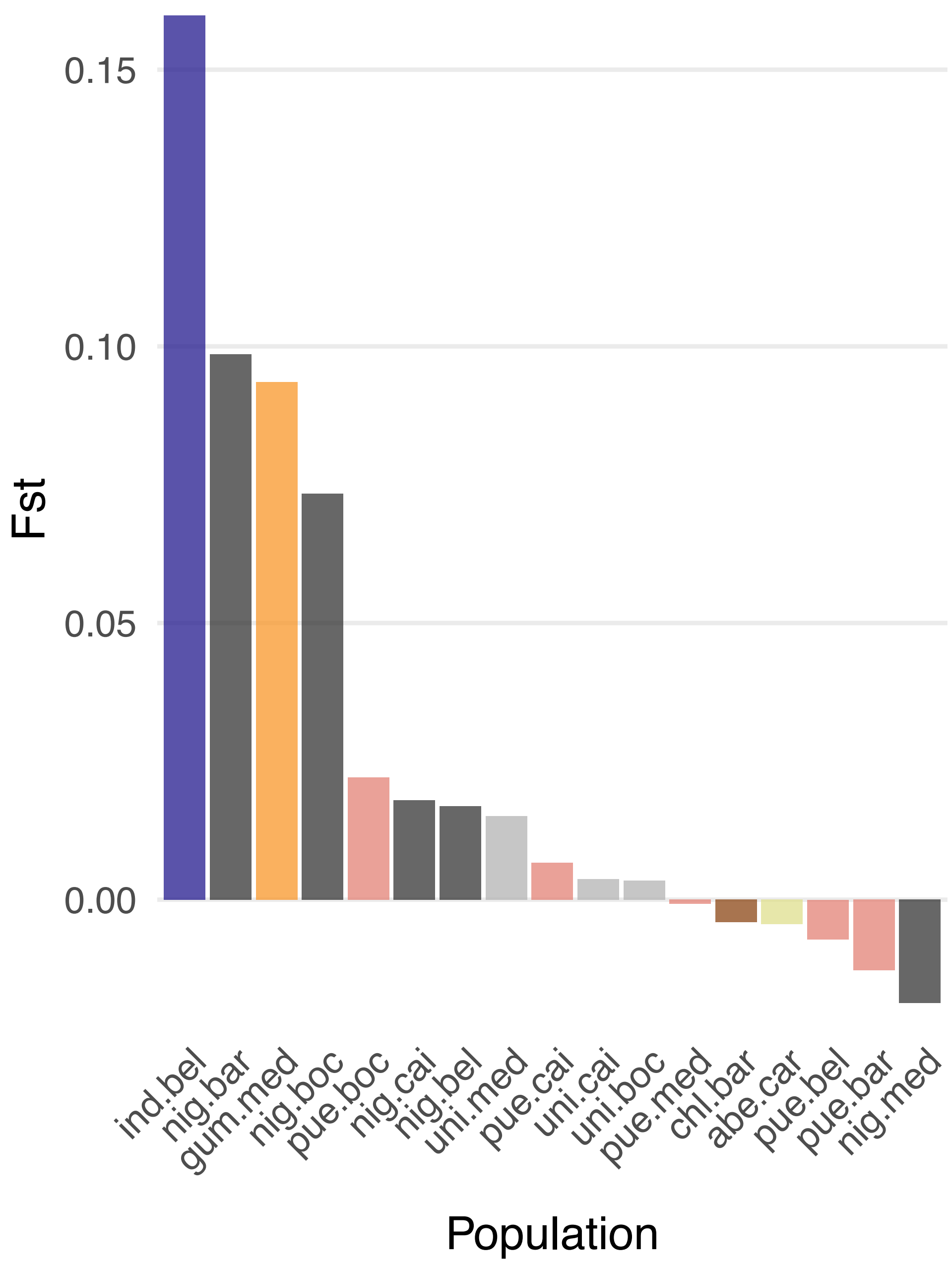
Visualize population structure

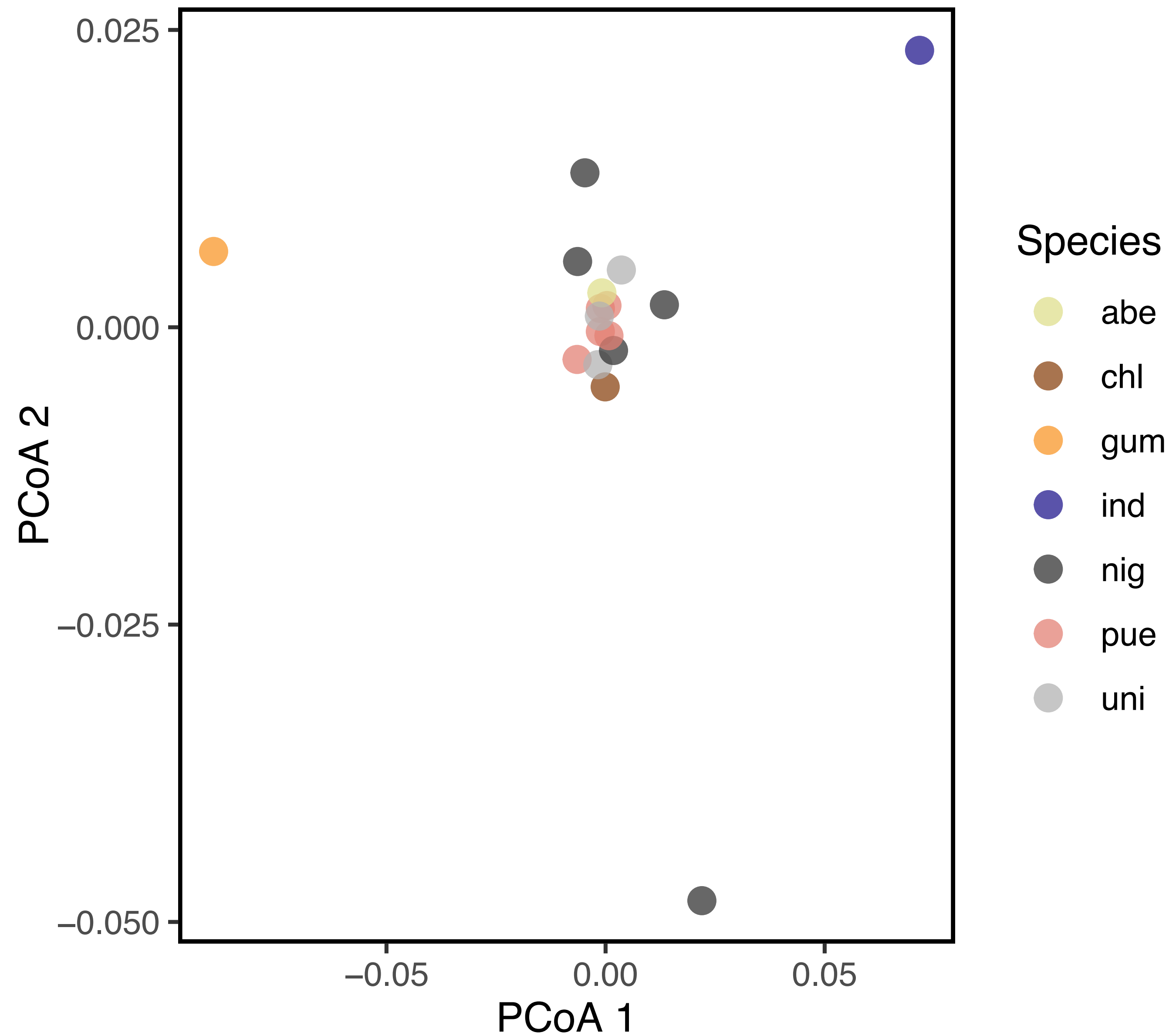
Exercise 3

- **Reduce high-dimensional genetic data** to 2–6 axes for visualization
  - Calculate pairwise distances and project samples or subpopulations into a coordinate system that preserves these distances (**PCoA,** principal coordinates analysis)
  - Find axes (principal components) that capture the maximum variance in genotype data (**PCA**, principal component analysis)
- Individuals or subpopulations **cluster by genetic similarity**, revealing structure across populations or species
- PCoA is suited for subpopulations and microsatellite data, while PCA can be used on the level of individuals and SNP genotypes

Visualize population structure

Exercises 2/3





Useful R functions for population genetics

Summary

Calculate *F*-statistics for all loci

```
pegas::Fst(as.loci(x))                                    # x = genind object
```

Calculate population-specific F_{ST}

```
betas(x)                                                  # hierfstat package
```

Visualize population structure using PCoA and pairwise F_{ST}

```
d <- genet.dist(x, method = "Nei87")                      # hierfstat package
p <- pcoa(as.matrix(d))
# … plot first two principal components as scatterplot using ggplot
```

New R concepts: tidyverse and ggplot

Summary

Tidyverse pipes

```
tb <- df %>%        # create object (copy)
   mutate() %>%     # 1st operation
   arrange() %>%    # 2nd operation
   select()         # 3rd operation
```

Basic ggplot syntax

```
ggplot(data = tb, aes(x = var1, y = var2)) +   # data structure (map variables)
   geom_bar() +     # geometric shapes representing the data
   labs() +         # set plot and axis labels
   guides() +       # customize plot legend
   theme()          # customize non-data, e.g. fonts, gridlines
```

Key takeaways

Summary

- ***F*-statistics** describe how much of the total genetic variation occurs within individuals, within populations, and between populations
- F_{ST} quantifies **genetic differentiation** (how much populations have diverged / degree of gene flow) by comparing within- vs. between-population allele frequency variation
- F_{ST} can be calculated **per population** to assess how distinct each population is, or calculated **pairwise** to visualize population structure (e.g. with PCoA)
- Analyzing **population structure may reveal effects of gene flow, selection, drift**, and informs conservation management

F_{IS} and self-fertilization in hamlets

Homework

Hamlets are simultaneous hermaphrodites with external fertilization. Discuss whether self-fertilization occurs regularly according to *F*-statistics.

- How would the number of heterozygotes be affected by self-fertilization?
- What would you expect for the inbreeding coefficient F_{IS} in this case?
- Use the *H. puella* dataset ("caribbean") to check your prediction

Course outline

Preliminary, may be subject to change

Class	Date	Topics	Script
01	Apr 10	Introduction and setup	01_intro.R
02	Apr 17	Hardy-Weinberg equilibrium	02_hwe.R
03	Apr 24	*F*-statistics and population structure I	03_popst.R
–	May 01	Labor Day	
04	May 08	Genome sequencing and assembly	04_asm.sh
–	May 15	Himmelfahrt Break	
05	May 22	Variant calling and SNPs	05_vcal.sh
06	May 29	Population genomics and genetic diversity	06_gdiv.sh
07	Jun 05	Population structure II	07_clust.sh
08	Jun 12	Selection	08_sel.R
–	Jun 19	Student presentations (no exercises)	
09	Jun 26	DNA barcoding	09_barcode.sh
10	Jul 03	Metabarcoding / eDNA	10_microb.txt
11	Jul 10	Metabarcoding / eDNA II (optional)	11_edna.R